Harsh Tripathi

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Professional Summary

Data-focused Python Developer and aspiring Data Analyst with experience in machine learning, automation, and Generative AI. Skilled in data analysis, predictive modeling, and building data-driven solutions using Python.

Experience

Python Developer Intern –Ishan Ceramics Private Limited, Agra (Oct. 2025 – Mar. 2026)

- Built **GenAI-powered automation solutions** using Python and LLMs to streamline workflows.
- Developed **AI agents** for data-driven decision-making and efficiency improvement.
- Created **AI-enhanced image processing pipelines** using OpenCV and NumPy.
- Integrated **LLM workflows with ERP systems and databases** for intelligent automation.
- Converted business needs into **GenAI-driven insights and solutions** using prompt engineering.

Full Stack Web Developer Intern – Infoseeds, Delhi (Jun. 2024 – Aug. 2024)

- Improved frontend performance by optimizing components for **responsiveness and cross-browser compatibility**.
- Conducted **UI testing and bug reporting**, ensuring high-quality user experience across applications..
- Validated and tested **REST API integrations**, ensuring correct data communication between frontend and backend.
- Collaborated with development teams to **identify and fix UI/UX issues**, improving usability.
- Assisted in improving overall application quality through **structured testing and debugging practices**.

Projects

Monkeypox Detection System (Machine Learning Project)

- Developed ML models using **Xception, EfficientNet, ResNet, MobileNet**.
- Applied **ensemble learning techniques** to improve prediction accuracy.
- Performed **data preprocessing, augmentation, and feature engineering**.
- Evaluated models using Accuracy, Precision, Recall, and F1-score.

Air Quality Index Prediction System

- Built predictive models using **Linear Regression, Random Forest, XGBoost, ANN**.
- Performed **EDA, data cleaning, and feature engineering**.
- Visualized insights using Matplotlib and Seaborn.
- Evaluated performance using **R^2, MAE, RMSE**.

Excel-ai-interviewer

- Built a **GenAI-based Excel interviewer** using LLMs for real-time conversational assessment.
- Developed **AI-driven question generation and adaptive interview flow**.
- Designed an **end-to-end AI interview system** with chat interface and insights.

Technical Skills & Tools

Programming: Python, SQL with strong experience in data processing and scripting.
Data Analysis: EDA, data cleaning, and visualization using Pandas, NumPy, Matplotlib, Seaborn.
Machine Learning: Regression, classification, clustering with model evaluation and tuning.
Libraries & Tools: Scikit-learn, TensorFlow, Keras, OpenCV, Jupyter Notebook.
Technologies: Power BI, Tableau, Excel, Git, REST APIs, MySQL, MongoDB, PostgreSQL.
AI Tools: Generative AI, GitHub Copilot, Google Gemini, OpenAI (ChatGPT, APIs), prompt engineering basics.

Education

B.Tech – Information Technology (CGPA: 6.4) (2021–2025)
Hindustan College of Science and Technology, Mathura

Senior Secondary Education (ISC Board) (2019–2020)
Ragendra Swarup Public School , Agra